

Artificial Intelligence in Clinical Decision Support

From Model Performance to Clinical Trust

Prof. Dr. Utku KÖSE

Süleyman Demirel University, Dept. of Computer Engineering

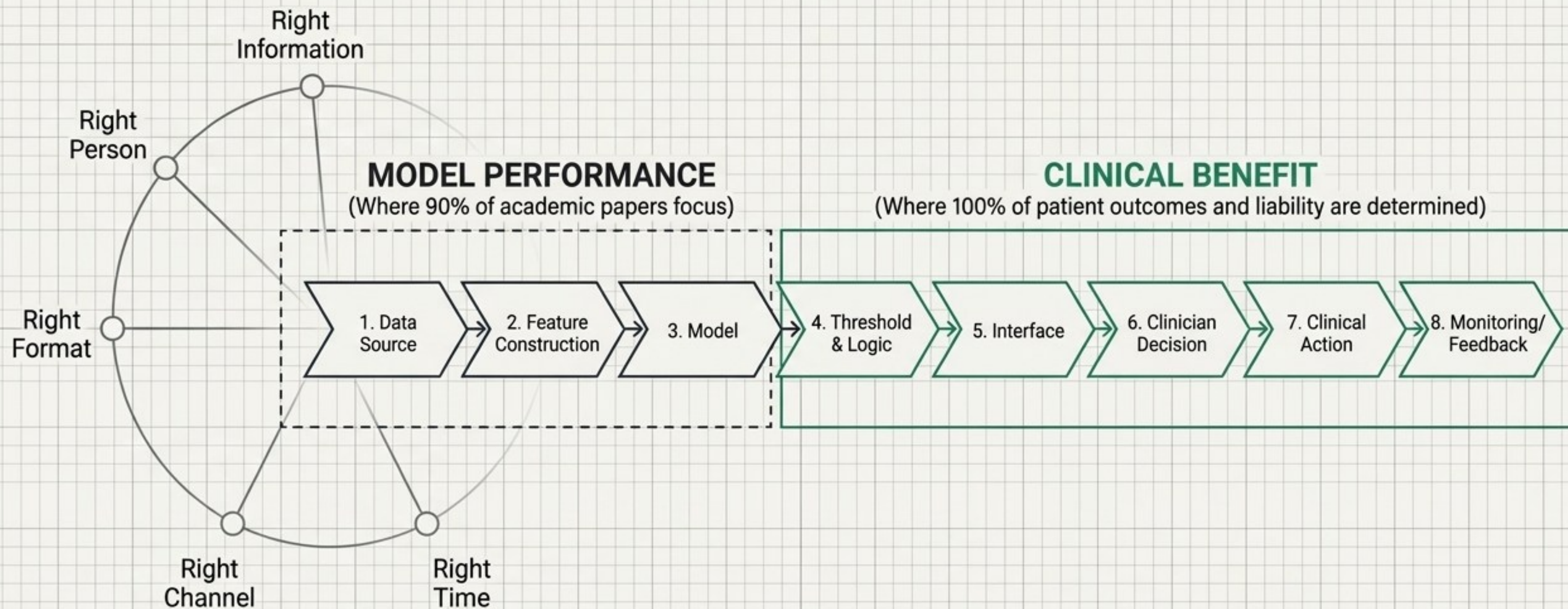
utkukose@sdu.edu.tr / utkukose@gmail.com / <https://www.utkukose.com>

A CDSS does not diagnose; it allocates clinical attention.

The system does not declare a condition is present; it proposes that a patient be prioritized. This distinction matters both clinically and legally.

THE RESPONSIBILITY IMPERATIVE

Literacy in health AI begins with responsibility, not technique. **The party that bears the cost of an error is never the party that wrote the code.**



KNOWLEDGE-BASED SYSTEMS

DATA-DRIVEN SYSTEMS

MECHANISM

Explicit rules derived from clinical guidelines.

Learned patterns from historical patient data.

CORE STRENGTH

Always readable, auditable, and straightforward to version.

Captures complex patterns no human could explicitly formulate.

CORE WEAKNESS

Rule bloat over time; fails to capture clinical nuance.

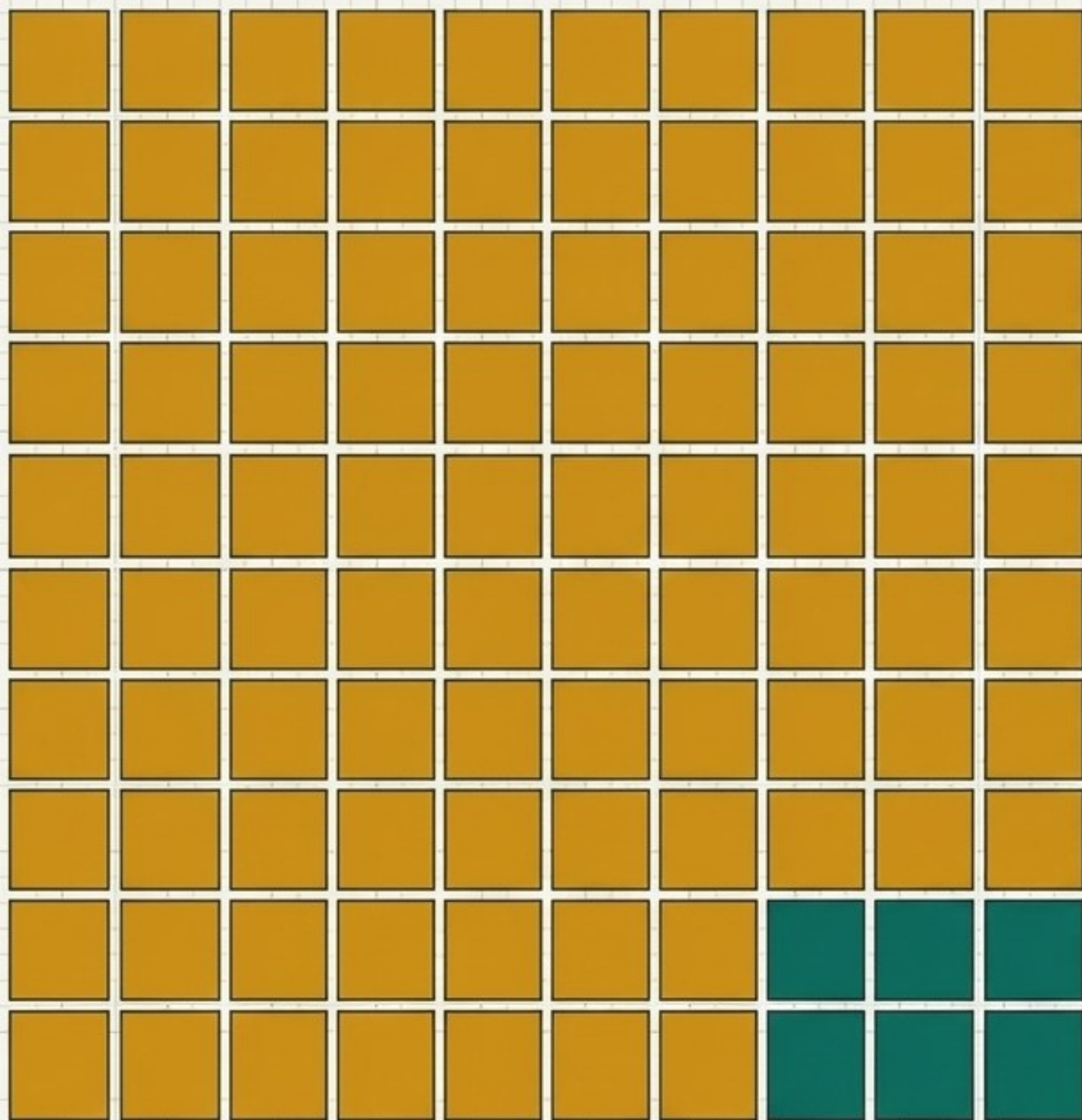
Shortcut learning: memorizing local infrastructure. Degrades upon transfer.

CLINICAL EXAMPLES

Drug interaction alerts, dose limits, allergy controls.

Sepsis risk scores, readmission prediction, lesion marking.

Hybrid Reality: Most modern field systems place a **learned data-driven score** behind a knowledge-based rule threshold.



The Illusion of Performance: The Epic Sepsis Case

Validation Parameters:

Michigan Medicine external validation
(38,455 hospitalizations).

The Claim: Vendor claimed AUC of 0.76–0.83.

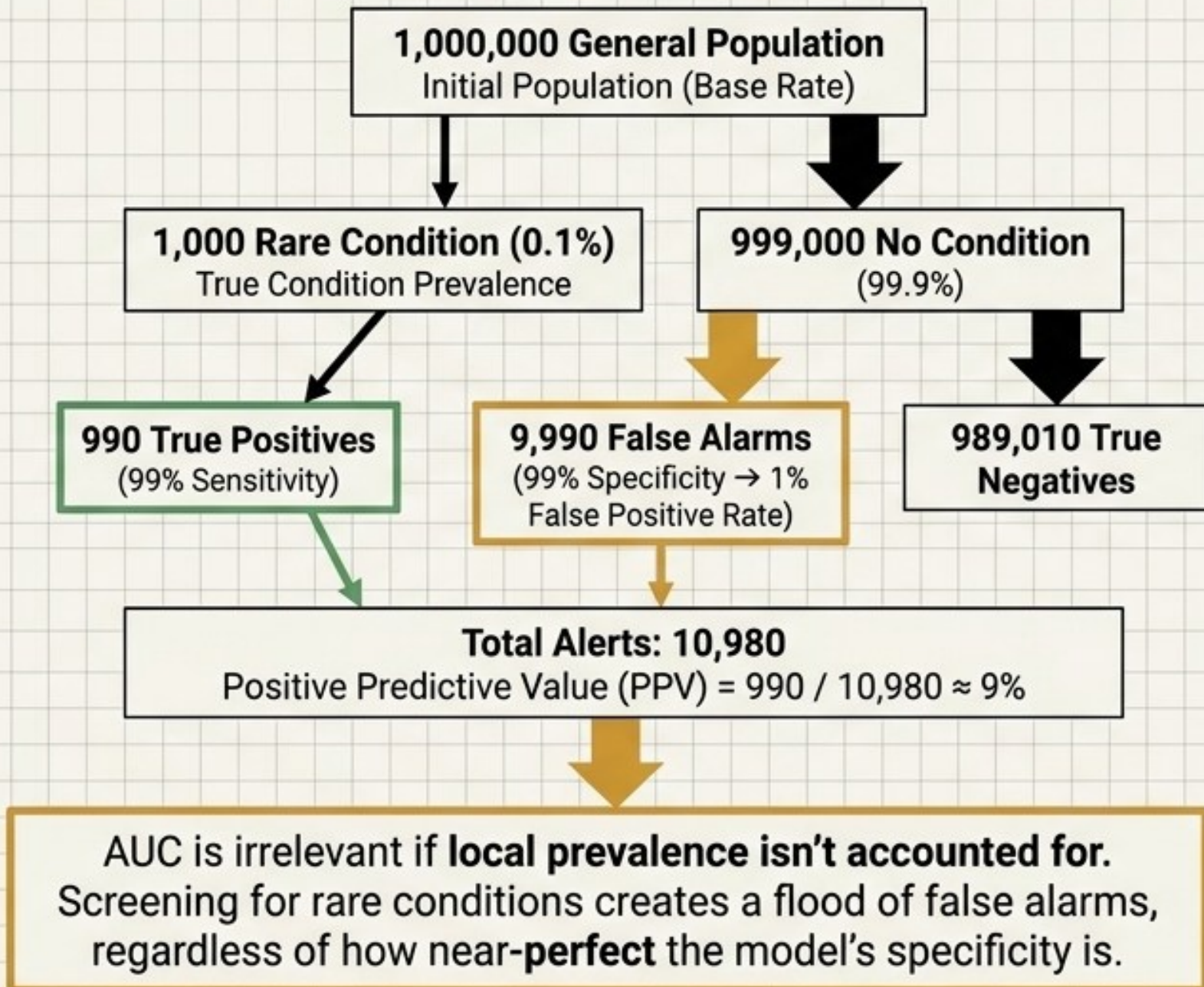
The Reality: Independent validation revealed an AUC of 0.63 and a Positive Predictive Value (PPV) of 12%.

The Clinical Impact

A **12% PPV** means that for every 100 alerts the system triggers, **88** result in a patient being reviewed without cause. High specificity (**83%**) masks an unmanageable false alarm rate on a shift roster.

The Prevalence Effect & Human Error

The Bayesian Trap



Alert Fatigue

Ignoring correct alerts because the system's baseline is noise.

Clinicians learn to dismiss alerts entirely unread.

Automation Bias

Ignoring contrary clinical signs because the AI is silent.

Clinicians falsely interpret system silence as medical reassurance.

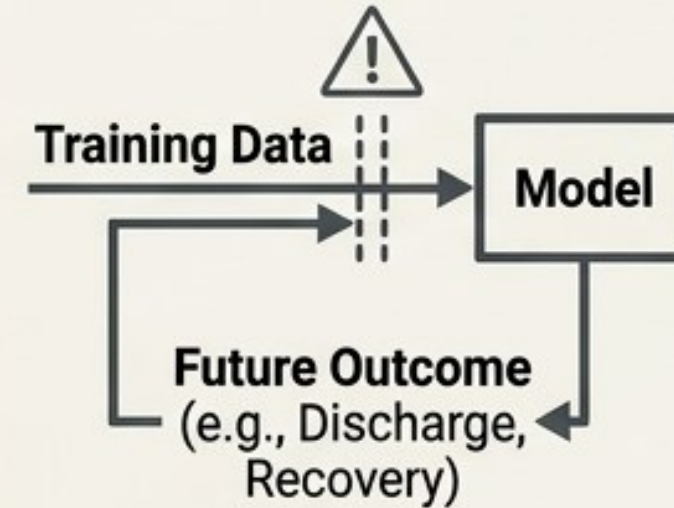
Four Modes of Silent Failure



1. Shortcut Learning

The model learns the device, not the disease.

(e.g., learning the marker a specific machine prints on an image rather than the pathology).



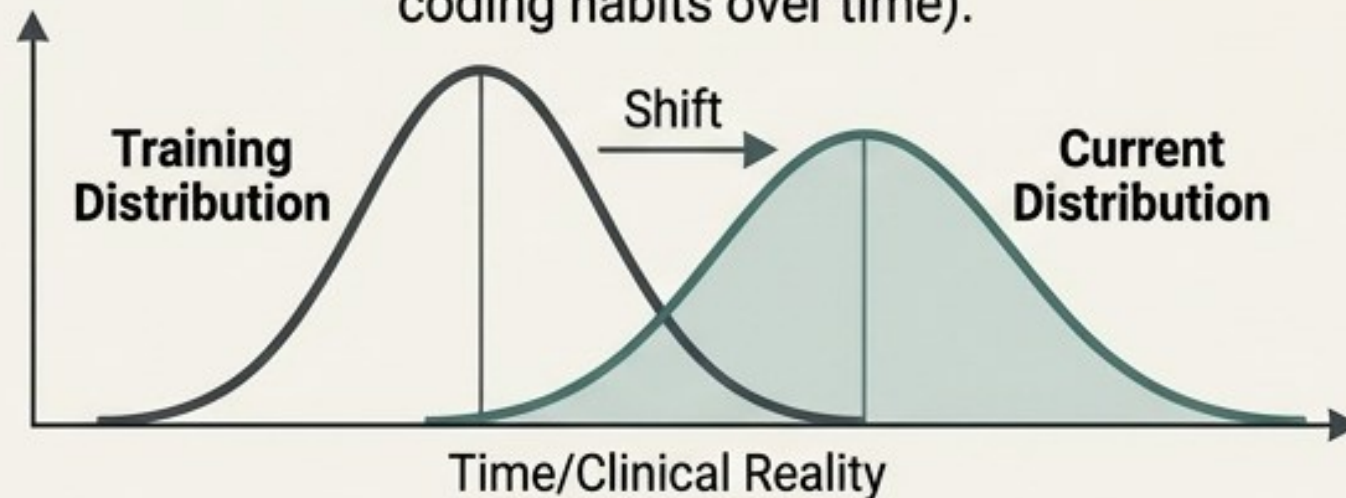
2. Target Leakage

Future data sneaks into the training.

(e.g., a consequence of the clinical outcome inadvertently entering the model as a predictive feature).

3. Distribution Shift

The model stays fixed, but clinical reality moves.
(e.g., changing patient profiles, protocols, or billing coding habits over time).



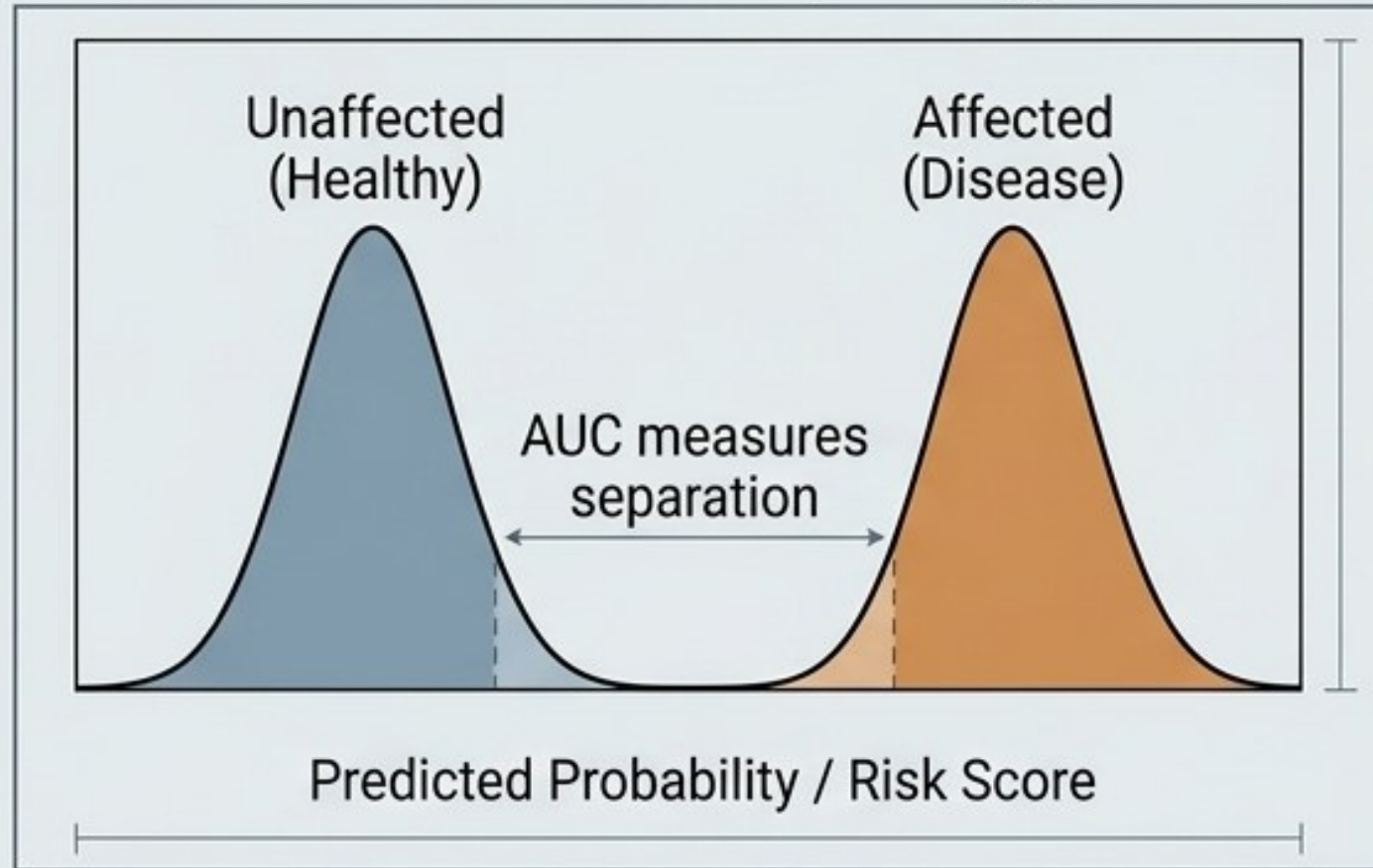
4. Subgroup Blindness

Acceptable overall accuracy masks localized harm.

(e.g., a high overall AUC hiding markedly lower performance in a specific demographic).

The Trust Metrics: Discrimination vs. Calibration

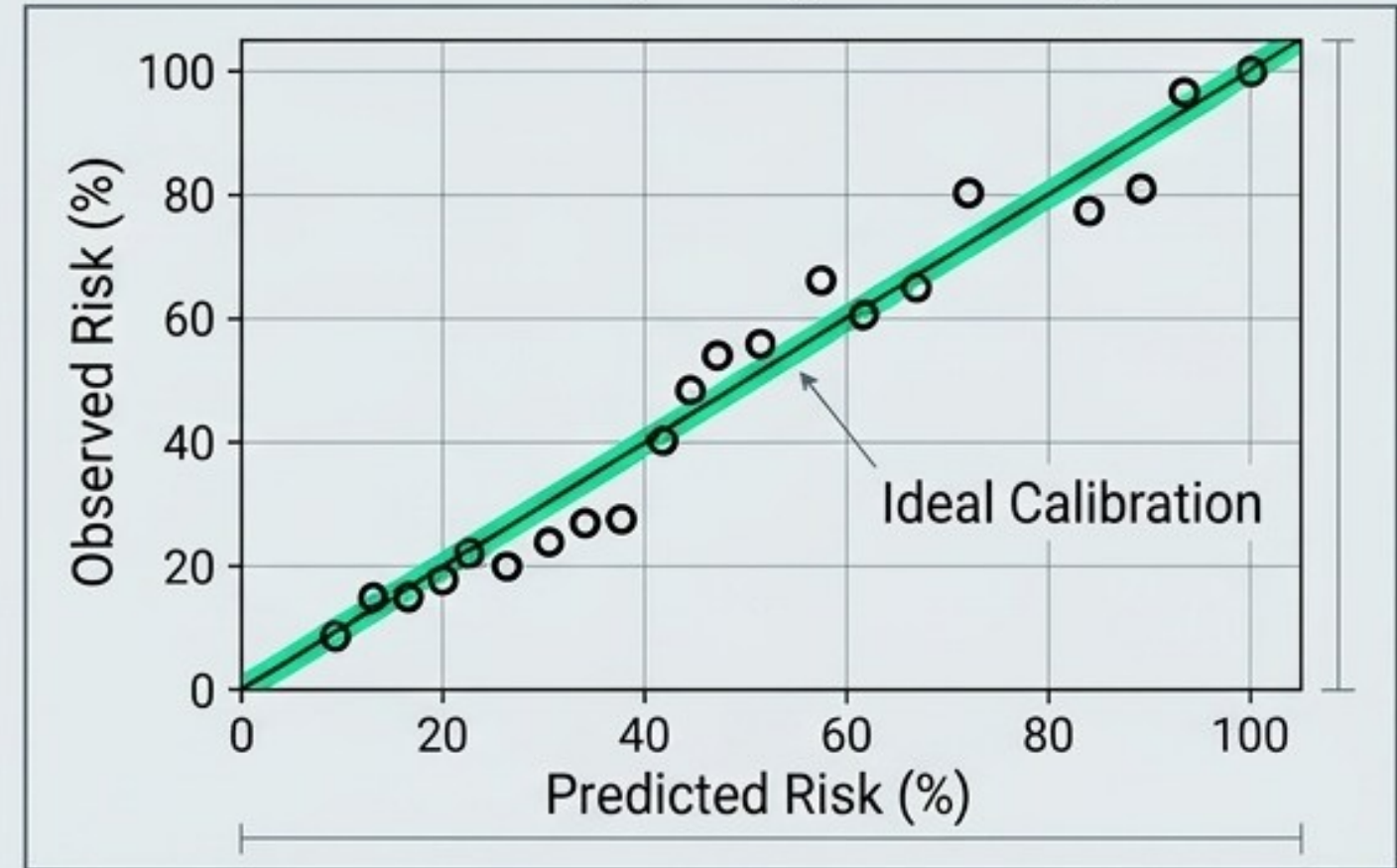
Discrimination (Ranking)



Discrimination (Ranking)

Measures the ability to separate an affected individual from an unaffected one. Metric: AUC.

Calibration (Reality Matching)



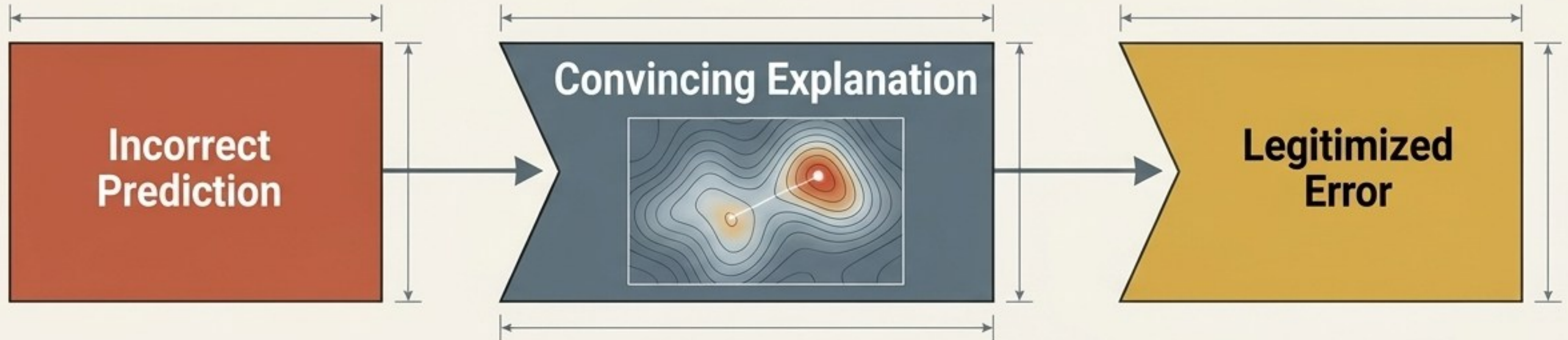
Calibration (Reality Matching)

Asks: If the model assigns a risk of 20%, do 20 out of 100 patients actually experience the event?

The Rule of Thresholds

Where a threshold is set for a clinical decision, Calibration takes absolute precedence over Discrimination.

Explainability is Not Correctness



Core Insight

Explainability does not carry a causal claim. An incorrect prediction can generate a perfectly logical-looking perfectly logical-looking explanation, tricking the clinician into legitimizing the error.

The True Purpose

Explainability's strongest contribution is error detection (finding data leakage or fairness issues), not proving the AI is right.

Regulatory Context

Under the US FDA 21st Century Cures Act, explainability is the threshold determining if software is regulated as a device—clinicians must independently review the recommendation basis.



Building for Trust: Data & Boundaries (Stages 1–5)

Stages 1-3: Moment, Data, Clean

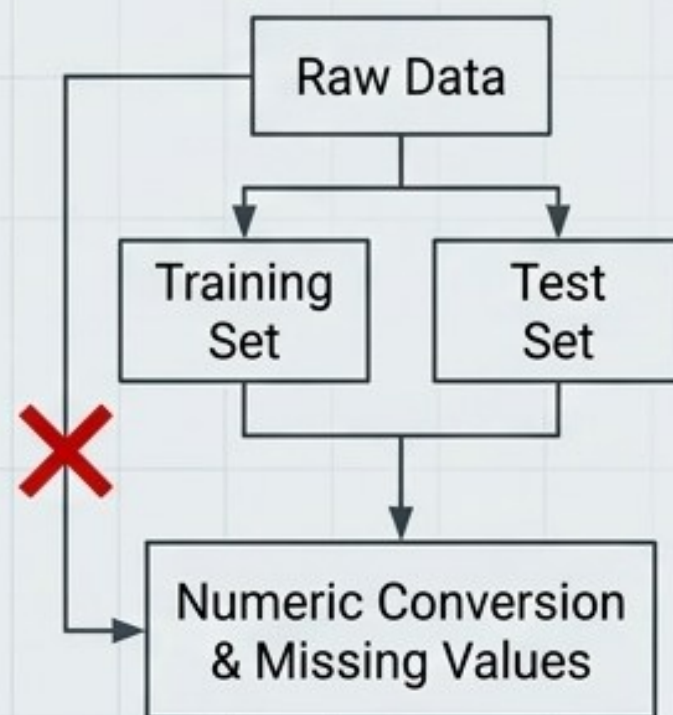


Define the exact decision moment (e.g., 6 hours post-admission). This draws the absolute boundary of allowable information.



Define the cohort explicitly (who is excluded matters as much as who is included).

Stages 4-5: Split & Convert (The Sequence of Safety)



Data must be split into training/test groups at the patient level **BEFORE** non-numeric conversions or filling missing values.

Reversing this order causes silent target leakage, lifting test results falsely while looking reasonable in the code.

Building for Trust: Models & Guardrails (Stages 6–11)

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Stage 7: Thresholds

The model must return probabilities, not labels. The clinical team—not the algorithm's default—sets the threshold based on the cost of false alarms vs. misses.

Stage 8: Measurement

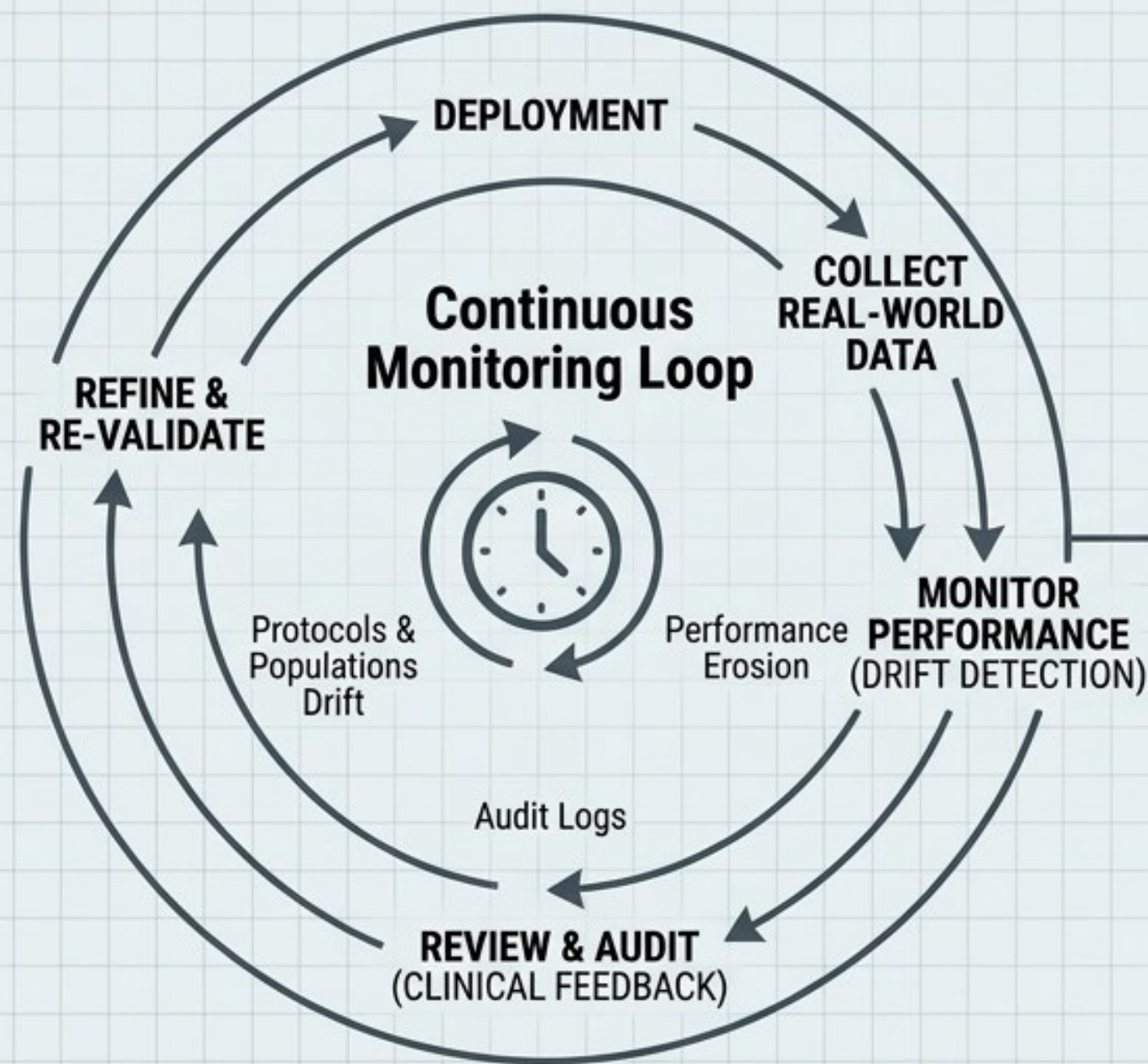
Compliance with reporting standards (TRIPOD+AI, DECIDE-AI).
Must report calibration, predictive values at the threshold, and subgroup breakdowns.

Stage 10: Safety Guardrails

The system must actively abstain from predicting when:

1. **Confidence is low** (sitting on the threshold line).
2. The patient is an **outlier** from the training population.
3. **Critical data is missing** (preventing predictions based on filled averages).

Deployment & The Software Lifecycle



Standards Mapping

IEC 62304 Class	Potential Harm	Documentation Rigor	Risk Management (ISO 14971)
Class A	No injury or damage to health	Minimal	Basic Risk Analysis
Class B	Non-serious injury	Moderate	Detailed Risk Analysis & Mitigation
Class C	Serious injury or death	High (Full Compliance)	Comprehensive Risk Management & Validation

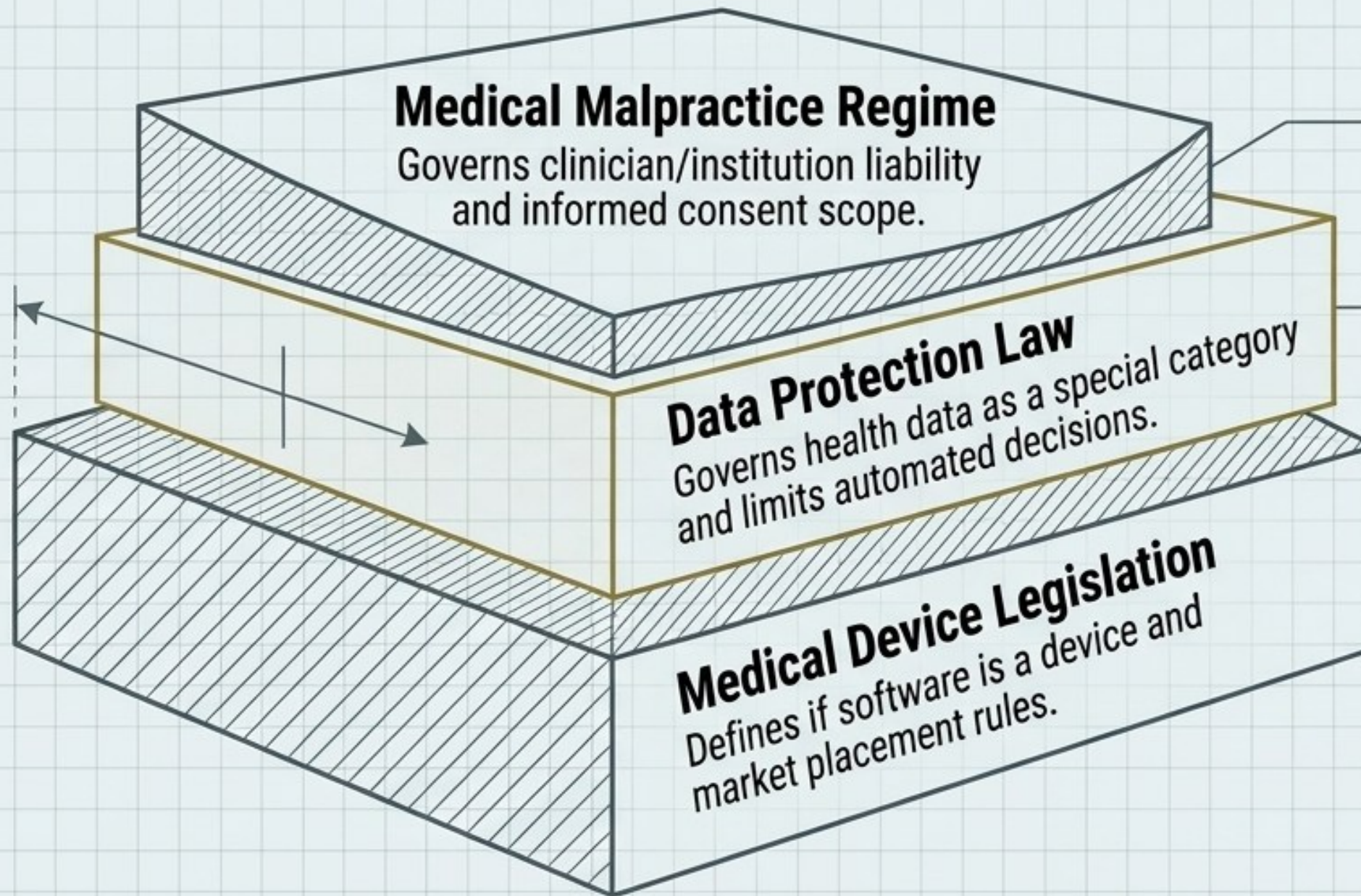
Stage 12: Interface Design

 Never display a single number. The decision, the state of the guardrails, and the reasoning must be presented together to enable independent clinical review.

Stage 13: Post-Deployment Monitoring

 Performance erodes silently as protocols and populations drift. Deployment without continuous monitoring approaches deployment without validation.

The Regulatory Framework & Liability



EU AI Act Rollout

Aug 2028

Embedded medical device AI.

Dec 2027

High-risk standalone systems.

Aug 2026

Transparency obligations active.



US FDA Stance

Lawful updating of learning systems requires a Predetermined Change Control Plan.

The GenAI Era: Coding is Cheap, Verification is Costly

Generative AI allows a clinician to build a working CDSS in hours without knowing Python. The burden entirely shifts from writing code to assessing code. **GenAI assumes no responsibility.**

Trap 1: Fabricated Thresholds

The LLM hallucinates clinical cut-off values with no traceable source, written in highly confident language.

Trap 2: Silent Leakage

The generated code inadvertently mixes test data into the learning process, creating falsely high performance.

Trap 3: Hallucinated Citations

References to non-existent validation studies—the most dangerous output because they appear formally correct.

The Clinical Assessor's Checklist



1. Did we validate externally? (A performance claim without independent external validation remains only a claim).



2. Did we check local calibration? (AUC is secondary; the model must match real-world probabilities and local disease prevalence).



3. Is the threshold our decision? (The cut-off must be a clinical decision based on false-alarm costs, not a model default).



4. Are the guardrails active? (The system must automatically abstain on missing data, low confidence, and outlier patients).



5. Can we review the reasoning? (The interface must allow independent clinician review to satisfy both safety and regulatory requirements).